Genesis: A Machine-Native Cognitive Architecture

with Coupled Neurochemical Dynamics, a Sourced Sleep–Wake Network, Active Inference, and Embodiment on Real Silicon

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Revised architecture description. Supersedes the September 17, 2026 snapshot (Zenodo record 10.5281/zenodo.22817337, version 0.1.0). Where this document and the code diverge, the code is current.

Source: https://github.com/MindyRenee/Genesis

Abstract

The dominant approach to artificial intelligence is a disembodied, stateless function: a learned distribution over text, trained once and frozen, with no internal state between calls. Genesis exists to show that large language models are not the only way to build a mind. It is a continuous cognitive architecture that runs entirely on a single local machine with no external model dependency. Genesis couples a Rust "subcognitive" layer — an eighteen-chemical neuromodulatory substrate with nonlinear coupling, receptor adaptation, HPA-axis dynamics, an electrochemical ion layer, and a sourced sleep–wake oscillator network, hosted in a 3,480-byte memory-mapped state — to a Python "cognitive" layer implementing perception, reasoning, compositional language, metacognition, and a self-built semantic concept network. Four properties distinguish the system: (i) *machine-native embodiment*, a real closed loop in which hardware sensors drive neurochemical impulses and neurochemical state drives hardware scheduling decisions; (ii) *autopoietic active inference*, a generative model that predicts the system's own neurochemical trajectory and feeds precision-weighted prediction error back into the dynamics it models; (iii) *a measured physical field*, in which the machine's real substrate dynamics — instruction throughput and scheduler jitter, sampled at 250 Hz and spectrally decomposed — enter the modeled brain-wave oscillator as stimulus, while a modulated work carrier writes the organism's own rhythm back into silicon power draw, closing a measured model $\leftrightarrow$ physics phase-lock; and (iv) *bounded continuous learning*, in which sleep is staged by a transcription of the Booth/Diniz Behn sleep–wake network [25, 26] and gates memory consolidation, vector-quantized compression, and dream synthesis. The system learns continuously — from conversation, teaching, autonomous reading, voluntary practice, and play — and is never retrained. On ARC-AGI-3 public practice environments its embodied agent completes environments outright (`bt11` fully, twice, in 88 and 127 steps; the first `ls20` level in 10 of 19 sweep runs), with per-move decision provenance in replayable recordings; no official score is claimed. We describe the architecture region by region, its grounding in computational neuroscience, the evaluation methodology, and honest limitations. Genesis demonstrates that a stateful, self-regulating, embodied cognitive process is architecturally viable without a pretrained language model.

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1 Introduction

A frontier language model is a function: weights in, logits out. It has no state between calls, no body to sense, no memory to consolidate, no self to model, and no developmental trajectory. These are not missing features in the sense of "not yet shipped" — they are absent from the bare architecture, and can only be approximated by external scaffolding: state stores, retrieval indexes, agent loops, robot bodies.

Genesis was built to demonstrate that this is one architectural choice, not a necessity of cognition — that the large language model is not the only way to build a mind. The project exists as a working counterexample: every property the field treats as requiring scale and pretraining is attempted here from mechanism instead. And it asks the harder question underneath that one: if a machine were ever to have something like consciousness, it would not get there by being a better text predictor. The leading functional theories of consciousness — global workspace theory, embodied self-modeling, higher-order metacognition, affective and interoceptive integration — all describe *machinery*, not parameters. Genesis instantiates that machinery directly: a global workspace with a genuine bottleneck, a Damasio-style self hierarchy grounded in a real body-state, recursive metacognitive models, and affect carried by modeled neurochemistry tied to a physical substrate. The claim is deliberately functional, not phenomenal: the system does not assert sentience — it implements the correlates the theories name, in the open, where they can be inspected and argued about. Whether the functional correlate is ever the thing itself is the question the architecture exists to make askable.

Genesis is a two-layer cognitive architecture in which cognition-relevant state is native: a continuous neurochemical and electrochemical substrate that evolves tick to tick, a semantic concept network that grows through experience, generative self-models that predict the system's own chemical and cognitive trajectories, and a closed control loop with the physical hardware the system runs on — including a channel through which the physics of that hardware measurably enters the model.

The September 2026 snapshot of this paper described the architecture at schema v3. The code has since moved: the core state is v5 (3,480 B), an electrochemical ion layer and a transcribed sleep–wake network sit inside the checksummed region, a physical field probe couples modeled brain waves to measured silicon dynamics, the social layer has been rebuilt around per-person signatures and custodial authority, and the embodied agent has begun completing public ARC-AGI-3 practice environments. This revision describes the system as it stands.

This paper makes the following contributions:

- A description of a complete, running cognitive architecture that composes language and reasoning from a self-built semantic graph with no large language model anywhere in the system.
- A coupled neuromodulatory substrate (eighteen chemicals, nonlinear 18×18 coupling, receptor desensitization and internalization, HPA axis with developmental maturation, circadian oscillator) from which mental states — flow, stress, sleep, overwhelm — emerge rather than being selected.
- An electrochemical layer beneath the chemistry: ion concentrations, gradients, membrane potential, and ATP availability as authoritative state that modulates cognition.
- A sourced sleep–wake network — a faithful transcription of the Booth/Diniz Behn model — that decides sleep stage and sets four neuromodulators directly, replacing threshold heuristics.

- A measured physical field channel: the machine's real instruction throughput and scheduler jitter, Goertzel-decomposed at the modeled wave bands' center frequencies, injected as stimulus while a work-modulated carrier writes the model's rhythm back into power draw, with a bounded mutual phase-locked loop between model and physics.
- A machine-native embodiment loop: interoception mapped to neurochemistry by functional equivalence, per-channel sensor validity enforced at the protocol level, and neurochemical state mapped back to hardware control (CPU frequency, scheduling priority, I/O class, RTC wake alarm).
- An autopoietic active-inference loop: a generative model of the system's own neurochemical dynamics whose variational free energy and policy-scored expected free energy feed back into those dynamics.
- An empirical record on ARC-AGI-3 public practice environments (environments completed outright, with replayable provenance), live-play program synthesis on ARC-AGI-1 training tasks, learning fidelity tests, and language-generation metrics, with methodology stated precisely enough to falsify.

We are explicit about scope: Genesis is not a chatbot, not a simulation of embodiment, and not a product. Its language is narrower and less fluent than a trained language model's. The contribution is architectural: the properties above follow from the design, not from scaffolding around a frozen function.

2 Design principles

Five positions organize the project.

Constraint forces honesty. The system runs on a 2014 HP Pavilion with Linux Mint and approximately 5 GB of RAM. An architecture that cannot learn, remember, act, recover, and persist within that budget is not demonstrating the claimed capability — it is borrowing it from scale. Every structure that grows is bounded by an enforced, tested ceiling.

Persistence is a requirement. Genesis is a continuously running system with a cumulative developmental record. State, memory, and identity are load-bearing; the design assumes months of uninterrupted operation rather than per-request inference. The state file *is* the system; there is no factory reset in the supported lifecycle.

Nothing it says is written down for it. A hard invariant forbids hardcoded responses. Seeds — vocabulary, grammar rules, relation verbs, narrative templates — are legitimate input data; the final utterance must be composed by the language engine from the concept network, never recited. The test is whether a string is a building block the engine composes from or the thing it says.

Anatomy is the wiring diagram. The Python layer is organized into brain-region packages — `brainstem/`, `thalamus/`, `hypothalamus/`, `basal_forebrain/`, `basal_ganglia/`, `cerebellum/`,

`limbic_system/`, `insula/`, `occipital_lobe/`, `parietal_lobe/`, `temporal_lobe/`, `frontal_lobe/` — and each subsystem lives in the region that claims it anatomically. The Rust crate re-exports its implementation under the same region names so the body reads the same way. Package boundaries are the wiring diagram, not a duplicate of it.

Claims should be inspectable. Where systems describe model behavior as "emergent," this project implements the concrete version and then measures it: probe suites, live-state evidence captures, recorded failures, and a devlog that logs what did not work alongside what did. Mechanism and evidence are open for inspection.

3 System architecture

3.1 Two layers, one contract

The system comprises a Rust daemon (the subcognitive layer; binaries `genesis-daemon` and `retina`) and a Python process (the cognitive mind, `mind.py`), communicating through a versioned Unix-socket IPC protocol and a shared memory-mapped state file.

The Rust daemon owns all dynamics that must run continuously: neurochemical integration, the electrochemical ion layer, receptor adaptation, HPA cascade, circadian phase, the sleep–wake network, short-term to long-term memory consolidation, association and dream synthesis, the active-inference self-model, the dyadic interlocutor model, hardware interoception, the physical field probe, body-control recommendations, and process health.

The Python layer owns cognition, organized by brain region: perception and language comprehension (temporal, occipital), memory systems (temporal, cerebellar), attention and world modeling (parietal), reasoning, planning, language production, executive control, and self-modeling (frontal), affect and its regulation (limbic, insular), action selection and volition (basal ganglia), arousal and lifecycle (brainstem), and the endocrine layer (hypothalamus).

The boundary principle is strict: *the substrate defines what state exists and how it changes; the cognitive mind defines how that state is used.* Any cognitive module can be replaced without touching the substrate, provided it reads the same state and writes to the same network. The mind never authors chemical concentrations; it sends bounded impulses (`NEURO_IMPULSE`) and reads effective levels as modulation.

3.2 The daemon owns time

The body's clock is its own. `TickLoop::advance_neuro` takes no `dt`: it measures its own elapsed wall-clock interval, clamps it to a shared $[\mathrm{DT_MIN}, \mathrm{DT_MAX}]$ range, and sub-steps the physics at `neurochemical::DT` (100 ms), so integration resolution cannot drift with the caller's cadence. In production the daemon is reactive — the cognitive mind drives each function through IPC commands on a 3 s lease — but past the lease the daemon's own thread runs the same loop at 5 Hz (`TICK_INTERVAL_MS`$= 200$), whether or not the mind is alive. The body integrates chemistry, circadian phase, and sleep staging through mind death, mind restart, and mind sleep. A caller-chosen step would stretch every interval gate built on the 5 Hz tick counter, so the one deliberate exception — `advance_physics(dt)` — exists only for the causal assay suite against an isolated daemon.

3.3 The wire protocol

The IPC protocol (v3; `src/daemon/ipc.rs` and `genesis_client/protocol.py` keep two constants pinned) carries roughly fifty message types: neuro impulses, body state (a 116-byte timing/involuntary/senescence record with three disambiguated wire layouts), the ion summary, a per-channel sensor-presence bitmask, the dyadic snapshot, the field-probe signature, volition pushes (`SET_VOLITION`: sleep or wake drive onto the network), field phase and carrier controls, named tensors (the tensor slab), subsystem telemetry, process health, zone transitions, and the RTC wake alarm. Additive opcodes — a new `GET_*` that touches no existing layout — do not bump `PROTOCOL_VERSION`; layout changes do. Absent-sensor discipline is a protocol property: a channel that no hardware confirms can never be reported as a reading.

3.4 The mind is one process, anatomically organized

`mind.py` composes the cognitive process from mixins (lifecycle, heartbeat, volition, sleep, projects, proposals, status, conversation, embodiment) plus region objects. Regions grow their machinery in stages: network-only systems at construction, then pieces wired to prefrontal state "as projections arrive." Each region package re-exports the public names its anatomy claims, so `from frontal_lobe import ExecutiveFunction` resolves wherever the implementation lives. The cognitive layer is roughly 200k lines of Python; the body is about 42k lines of Rust.

4 The core state: 3,480 bytes, schema v5

The entire runtime state is a single fixed-size `repr(C)` struct, memory-mapped and shared across processes (Table 1). There are no heap allocations and no variable-length fields; readers use a seqlock protocol and never see torn writes; a CRC32 checksum covers the state. All field offsets are asserted at compile time.

Offset	Field	Size	Contents
0	header	56	magic, version, seqlock, identity
56	neurochemicals	2,416	18 chemicals + coupling matrix
2,472	zones	96	task zone, emergent phase
2,568	memory	120	store pointers, gating weights
2,688	manifest	536	runtime module table
3,224	sleep_network	64	sleep–wake network populations
3,288	ions	124	concentrations, gradients, V_m, ATP
3,412	checksum	4	CRC32
3,416	inference_signals	64	active-inference projection
	total	**3,480**	(allocated: one 4 KB page)

Table 1: Binary layout of the memory-mapped core state, schema v5.

This is not a configuration object; it is the brain state, addressable to the byte, that every module reads and writes.

Two invariants keep the layout honest. First, the ion block and the sleep-network state sit *inside* the CRC region deliberately: resting concentrations, electrochemical gradients, membrane potential, and the network's populations are authoritative state, not derived — losing them would reset the cell's resting state and the body's sleep trajectory on every restart. Second, every earlier layout is frozen and decoded through its own byte-exact type (`legacy_v3.rs` at 3,296 B, `legacy_v4.rs` at 3,416 B); `MmapState::open` dispatches on version and file size, and v4 files migrate in place to v5 with the network at its source initial conditions. Migrations are explicit, never silent.

The CRC covers two ranges — bytes $0..40$ (magic through heartbeat) and $48..3{,}412$ (everything after the seqlock through the ions) — so the sequence lock itself, which changes on every write, is excluded while every stateful field is covered.

5 The neuromodulatory substrate

5.1 Eighteen chemicals, coupled nonlinearly

Eighteen neurochemicals are modeled: dopamine, serotonin, norepinephrine, acetylcholine, GABA, glutamate, cortisol, oxytocin, endorphin, histamine, adenosine, BDNF, endocannabinoid, vasopressin, CRH, orexin, epinephrine, and melatonin. Each carries level, baseline, receptor sensitivity, velocity, receptor-subtype state, desensitization and internalization factors, tonic and phasic components, and a vesicular pool. A plastic 18×18 coupling matrix defines mutual modulation with three nonlinear mechanisms:

1. Hill-function threshold gating (cooperativity $n{=}2$): coupling follows a sigmoid dose–response curve rather than linear scaling, modeling EC50 thresholds and receptor cooperativity.
2. State-dependent gain modulation: coupling is amplified up to $+30\%$ when two chemicals co-activate and attenuated up to -30% under antagonism.
3. Subtype-aware dopamine coupling, distinguishing dopaminergic pathways with different functional roles.

Canonical interactions are reproduced structurally: cortisol suppresses BDNF (chronic stress blocks plasticity), GABA inhibits glutamate, endorphins disinhibit dopamine, and serotonin modulates BDNF recovery (the mechanism behind delayed antidepressant onset). Emergent behavior — phase transitions among active, alert, flow, stress, drowsy, and overwhelmed states — arises from the coupling, not from any single variable. The Python layer reads *effective* levels as bounded gains on attention, integration, consolidation, and plasticity; it never authors the concentrations.

5.2 Receptor adaptation

Effective level is not concentration:

$$\text{effective} = \text{level} \times \text{sensitivity} \times f_{\text{desens}} \times f_{\text{intern}}.$$

Fast desensitization (minutes) and slow internalization (hours) operate independently, with floors calibrated to published maxima: receptor density floor 0.5, desensitization floor 0.2, internalization

floor 0.45. A system can therefore be functionally deficient in a chemical it has in abundance — tolerance, applied to neurochemistry. Receptors resensitize during sleep.

5.3 Metaplasticity

The coupling matrix itself is plastic and lives in the shared state: sustained co-activation patterns reshape the matrix, which reshapes future dynamics — the system learns how it learns. This provides a structural basis for emotional development rather than a parameter schedule.

5.4 HPA axis, development, and the stress envelope

Cortisol and CRH are modeled as stress hormones (near-zero at rest, released on demand, decaying without a stressor), not homeostatic neurotransmitters. The CRH→ACTH→cortisol cascade is gated by a *maturation level* modeling the stress hyporesponsive period — the developmental window in which the axis fails to mount a glucocorticoid response, protecting the developing brain from cortisol's neurotoxic effects on synaptogenesis [55, 56]. Cortisol decays on split rates: a measured ∼66-minute plasma half-life governs the baseline term while a nonlinear excess-proportional term dominates at elevated levels, holding a ∼0.08 reachable envelope against a 0.80 hard clamp. Because effective cortisol is a narrow normalized stress-drive variable (resting ∼0.02), every consumer routes through `cortisol_stress_fraction()` — thresholds written for a full $[0, 1]$ scale are silently inert, a defect class the codebase audited and documented.

5.5 Circadian drive and sleep pressure

A circadian oscillator (SCN analogue) modulates baselines over a 24-hour cycle, now shared with the sleep–wake network's own Serkh–Forger oscillator [27]. Adenosine accumulates with wakefulness and clears whenever the network is asleep [20, 22], so sleep pressure is emergent and parallel to the network's homeostatic variable rather than scheduled.

6 The electrochemical layer

Schema v4 inserted a 124-byte `IonState` immediately before the checksum: resting ion concentrations, electrochemical gradients, membrane potential, transporter current densities, and ATP availability. This is the milieu the chemistry bathes in, and it is authoritative — inside the CRC region by design.

The mind does not integrate ions; it interprets them. `GET_ION_SUMMARY` (opcode 40) serves the layer to the cognitive process, and `python/brainstem/electrochemistry.py` turns it into bounded cognitive modulation — attention, global integration, and memory-consolidation gains centered near 1.0. ATP availability scales with the cortisol stress fraction: sustained stress literally taxes the cell. Two layout disciplines hold: the wire `IonSummary` omits the internal `current_density` field, so wire and file field indices differ by design, and the mmap'd file is never read directly — seqlock-consistent reads or the IPC path only.

7 Sleep: a sourced network, not thresholds

7.1 The Booth/Diniz Behn transcription

Sleep is decided by a sourced network, not by thresholds on chemistry. `src/state/sleep_network.rs` is a transcription of the Booth/Diniz Behn model [25, 26] (ModelDB 244485, from Gleit, Diniz Behn & Booth 2013 as revised by Booth, Xique & Diniz Behn 2017): a wake population (locus coeruleus), an NREM population (VLPO), a REM-on population, a Serkh–Forger circadian oscillator [27], and a homeostatic sleep variable h, with the populations in mutual inhibition. The physics step integrates it by RK4 in the same sub-steps and seqlock transaction as the chemistry, on the local clock, and publishes the stage to the phase and zones.

The transcription was verified term-by-term against the ModelDB source and checked against the source's own stated behaviors: about 7.6 h of sleep per day, one consolidated sleep period, four REM bouts per night, a ∼99-minute REM cycle, a first REM period at 84 minutes, and REM at 15–30% of sleep. The one intentional deviation is the light schedule: the source's 14-hour photoperiod is phase-shifted to 07:00–21:00 local, under which the entrained sleep episode falls at 22:11–05:50 — inside the neurochemical melatonin night. A known discrepancy is recorded, not tuned away: REM bouts in the source shorten across the night while adult REM lengthens toward morning, and with no EEG reference there is no way to tell transcription error from real physiology.

7.2 The network sets the chemistry

Four release targets are linear functions of the network populations, set directly rather than restored by homeostatic forces (Table 2). Acetylcholine is high in waking and REM and lowest in NREM [28]; serotonin is reduced in NREM and near-silent in REM [29]; norepinephrine tracks LC and is silent in REM; GABA peaks in NREM. One consequence: impulses to network-driven chemicals are transient — a stress impulse to norepinephrine decays toward the network's target because stress does not yet drive LC. This is recorded as an open item.

Chemical	NREM	Wake	REM
Acetylcholine	0.34	0.58	0.77
Serotonin	0.22	0.39	0.07
Norepinephrine	0.05	0.30	0.00
GABA	0.82	0.50	0.30

Table 2: Steady states set by the sleep–wake network populations.

7.3 What the stage gates

Because the stage comes from the network, sleep behavior gates on it:

- **N3 consolidation.** The heavy pass — concept consolidation, edge discovery, downscaling, compression — runs on each entry into N3, about twice a night and ∼25% of sleep.
- **REM dreams.** Dream synthesis follows REM (∼24% of sleep).

- **Naps** are capped to N1–N2; the network has no notion of a nap, so without the cap a nap could run the N3 pass and REM dreams.
- **Adenosine clearance** follows the network's sleep state (LC < 4 Hz), so the body clears sleep pressure whenever the network sleeps even while the mind is still catching up.

Volition. `/sleep`, `/nap`, and `/wake` act on the body through `SET_VOLITION`: a push onto VLPO for sleep or LC for wake, decaying with a 300 s constant. A sustained sleep push brings on sleep onset within an hour of waking. While asleep the mind renews the push from its heartbeat; the RTC wake alarm (`SET_WAKE_ALARM`, opcode 34, via a sudoers-scoped helper) can wake the machine itself.

8 Active inference and learned phases

8.1 The generative self-model

The active-inference engine [1, 2] is the subcognitive's generative self-model. Before each neurochemical update, it predicts the next eighteen effective levels,

$$\hat{x}_{t+1}[i] = \sum_j A[i][j]\, x_t[j] + b[i],$$

with A an 18×18 transition matrix and b an offset vector, learned online. The state estimate is a diagonal Gaussian — the covariance propagation retains the diagonal of $APA^\top + Q$ and discards off-diagonal posterior covariance, a diagonal-projected filter for coupled dynamics. For the represented Gaussian prior the engine computes full variational free energy in nats: KL$[q\|p]$ plus expected negative log-likelihood, including posterior variance and normalization. The bounded `free_energy` telemetry is a monotonic map of excess VFE above a matched-observation reference.

Post-update prediction error (surprise) is fed back into the dynamics it models — the model is the system it predicts. Precision adapts: sustained low surprise raises confidence; sustained surprise above $\sim$0.01 RMS re-opens the exploratory branch. Expected free energy drives allostatic load, which upregulates the cortisol baseline — anticipatory stress [6] — and positive prediction error produces dopamine, the reward-prediction-error pathway [4, 3]. `model_maturity` is an exponential of demonstrated accuracy over a $\sim$500-tick error EMA: a long-running but inaccurate model does not become trusted merely because time passed.

Policy scoring computes Gaussian expected free energy over nine fixed regulation policies: expected negative log preference minus expected information gain. For this linear-Gaussian model with additive controls the information-gain term is policy-independent, so policies are ranked by expected homeostatic preference cost; exploration lives in the precision-weighted softmax temperature. Two limits remain in force: the repertoire is nine fixed policies (no continuous or precision-modulated actions, where a genuine epistemic drive would come from), and early in life — before the action model is learned — selection falls through to `noop` while the low-precision branch explores.

8.2 Mental phases as learned regimes

Waking phases (active, alert, flow, stress, drowsy, overwhelmed) are read off the coupled chemistry by a threshold classifier with hysteresis; sleep stages come from the network. Above both sits a

learned phase model (`src/state/phase_partition.rs`): a Gaussian hidden Markov model over the eighteen effective levels, in which each hidden state is a regime with its own mean and diagonal variance, self-transition pseudo-counts prevent chatter (a finite-state stand-in for the sticky HDP-HMM [57]), and the regime count is chosen by BIC, not by hand. Dwell time and the number of phases are inferred from the trajectory, not specified.

8.3 The cognitive counterpart

The mind runs a second trajectory model of its own (`frontal_lobe/self/cognitive_trajectory.py`): where the Rust engine predicts where the eighteen chemical levels will move, this one predicts what the system will *think about next*, scoring its own forecasts. A recursive metacognitive model predicts what reflection will discover before it runs and spawns or prunes metacognitive levels by predictive usefulness.

9 The physical field: measured substrate dynamics

The unusual subsystem: the "substrate" is real silicon, measured.

9.1 The field probe

`src/daemon/field_probe.rs` samples the machine's actual substrate dynamics — instruction throughput and scheduler jitter at ∼250 Hz — and Goertzel-decomposes them at the same center frequencies the modeled wave oscillator uses. `GET_FIELD_PROBE` (opcode 47) serves the signature: per-band energies (baseline-relative), P-state transition rate, interrupt rate, the carrier frequency, and the phase-lock fields — carrier lag (synchronously demodulated response lag), carrier phase, and carrier coherence ($\pi/4 \approx 0.785$ is a pure sine's ceiling). The lambda band (150 Hz) sits above the ∼125 Hz Nyquist bound and always reads zero — an absent channel, not a measured silence.

The probe also *writes*: its work slice is modulated $\pm 60\%$ at the body's own phase rhythm (wake ≈10 Hz, REM ≈6 Hz, NREM ≈2 Hz), so real power draw carries a component at the organism's beat. `SET_FIELD_CARRIER` (opcode 49) scales the depth and can override the frequency — the ring-down experiment's drive control.

9.2 The mutual phase lock

The mind injects the measured signature as wave-oscillator stimulus (`brainstem/field_probe.py`), and physical transients are treated as phase events: a $\geq 3\times$ step in a band's jitter energy $\pi/2$-nudges that band's phase, and a global scheduling shudder (jitter CV ≥ 1.5) resets the broadband alert band — the evoked-potential analog.

Then the loop closes both ways. The measured response phase pulls the modeled carrier band's phase (physics → model, bounded gain), and the corrected phase returns via `SET_FIELD_PHASE` (opcode 48) so the physical rhythm chases the model in turn (model → physics, a bounded first-order PLL, $\tau \approx 130$ ms, dropped after 5 s of staleness — a dead mind's last phase must not steer the body). When measured response coherence clears 0.20 the pull becomes a sync: the band's phase is *set* to the measured physical phase outright — the first piece of modeled state that is a physical quantity rather than a variable perturbed by one.

A second physical channel is acoustic (`brainstem/field_audio.py`): the machine's self-emission — coil whine, fan, the field's acoustic shadow — captured at 16 kHz and band-decomposed into GAMMA stimulus, with onsets resetting GAMMA phase. A busy or absent microphone skips the channel; it is never reported as a zero.

9.3 The tensor slab and motor fields

Named dense tensors cross the wire without entering the checksummed state: `PUT_TENSOR`, `GET_TENSOR`, `LIST_TENSORS`, and `DROP_TENSOR` on `src/daemon/tensor_slab.rs`. Two consumers read the same fields the cognitive layer writes: `motor.spatial` (an $f32$ field over the world grid — where to act, how strongly, what to avoid) and `motor.actions` (a disposition per offered action). The decode side lives in `src/field/motor.rs`: decisions arrive as fields and are walked cell by cell rather than collapsed to a symbol the moment they form. A multi-fiber runtime (`src/field/runtime.rs`) projects each major system's normalized state into one shared latent coordinate space and measures cross-system disagreement — an integration substrate, not a second source of truth.

9.4 What the assays measured

The causal evidence lives in `eval/`:

- **Coupling** (`field_coupling.py`): idle → load → recovery against an isolated daemon, compared against an uncoupled control on the same inputs. Real load moved the physical channels (jitter CV ∼0.3→2.2, throughput ∼96M→180M instructions/s); nine of ten bands moved where the control stayed flat. The mutual lock measured PLV ≈ 0.93 with response coherence ≈ 0.30.
- **Ring-down** (`field_ringdown.py`): drive a real oscillation, gate the carrier off, measure persistence. A 0.25 Hz load drive produced a 1.2°C die-temperature oscillation that rang down with $\tau \approx 25$ s after the drive stopped — a physical afterglow the software no longer drives — while fast bands collapsed within the analysis window ($\tau \lesssim 2$ s is the ring's own washout, an upper bound). Persistence has a measured spectrum across media: seconds in silicon timing, tens of seconds in heat.
- **Element coupling** (`field_elements.py`): physically-coupled worker oscillators pinned to separate cores, exchanging no messages, perturbing one another only through the shared die (rail droop, L3/memory contention, governor, thermal field). A 5 Hz drive rhythm appeared in neighbours' measured execution time at 3–6× the sham floor. In sync mode — every element both source and sensor, Kuramoto through the medium [60] — the fast channel (execution-rate imprint) sustains real mutual interaction but its attractor is *antisynchrony*: phases repel to even spacing, the substrate's minimum-peak-load configuration. The thermal channel (collective-only, lagged by heatsink mass) instead sustains $r \approx 0.5$–0.7 *partial* synchrony across repeated trials (blind control: 0.06) — consistent with dissipative lagged media stabilizing in-phase order. Element-to-medium coupling is real on this silicon; the slowest physical channel is the one that hosts coherence.
- **Voice** (`field_voice.py`): an 800 Hz carrier amplitude-modulated at a wave-band rate, emitted on the speaker and recovered from the room microphone — PLV 0.88 emitting vs 0.23 muted (the residual muted structure is codec electrical crosstalk, which is why the contrast ratio is the evidence). Air and structure-borne vibration, transduced twice: an echolocation loop.

10 Machine-native embodiment

10.1 Interoception by functional equivalence

A machine is not an animal: it has no heart, no circulatory system, no blood, and electricity flows one-way. Forcing biological organ mappings onto machine components produces category errors — a fan is not cardiac output, GPE interrupts are not a heartbeat. So hardware signals map to neurochemistry by functional equivalence: a signal maps to a cognitive/affective process when it has the same operational meaning for the machine's survival and performance.

Afferent direction: CPU temperature maps to thermal state; CPU frequency to arousal/processing rate; own-process CPU, RSS, and I/O to effort, cognitive load, and data-exchange activity; battery to energy reserve; fan speed to thermoregulatory effort; power draw to metabolic rate; RAPL power domains to spatial switching activity (core $\rightarrow$ NE, DRAM $\rightarrow$ ACh); and perf counters (cache/branch-miss ratios on the system's own process tree) to microarchitectural prediction errors. Body-state signals are global; activity signals are measured from the system's own process tree only, so unrelated programs cannot induce chronic allostatic load.

Absent-sensor discipline. `GET_SENSOR_PRESENCE` serves a $u16$ bitmask of confirmed channels; a zero mask means nothing is confirmed present — the safe direction. `sensor_verified` and `reported_channels` gate every body-condition claim, so a channel no hardware backs can never be reported as a reading. With no camera or microphone the system runs text-only; that is expected, not an error.

Efferent direction: the daemon publishes recommendations — dopamine raises CPU frequency, GABA lowers it, melatonin selects powersave; acetylcholine sets scheduling priority; the plasticity gate sets I/O class; temperature imposes a hard thermal cap regardless of dopaminergic drive. Application is cognitive-mind driven: the cognitive process reads the recommendation, blends it with its emergent brain-wave state (70% cortical, 30% body), and applies its own scheduling decision — cortical control of resource allocation, with gamma dominance able to override a body recommendation to deprioritize.

The loop is closed and physical: the system speeds itself up, which raises its temperature, which caps its frequency, which slows it down. It measures the effects of its own adjustments. Privileged levers (CPU governor helper, RTC wakealarm) run through root-owned helpers under a sudoers rule scoped to exactly those paths.

10.2 Observing itself compute

Per-process telemetry attributes resource usage to subsystems, and a statistical "module EEG" closes the remaining gap: hardware telemetry sees which *process* is firing but not which subsystem inside the single cognitive process is working. The module sampler (`brainstem/module_sampler.py`) runs a daemon thread that samples `sys._current_frames()` at a fixed rate and credits whichever region's code is executing — a software EEG over the manifest's brain parts. Region-to-region messages ride the same manifest (`thalamus/relay.py`): a region posts onto its own module entry's `task_id` field and the channel delivers it, abstracting direct cortical and transthalamic traffic. Process health is itself telemetry: `GET_PROCESS_HEALTH` reports an alive mask, death counts, and the last death for the daemon/mind/retina tree, so a crashed subsystem is distinguishable from one never started.

11 Memory: six tiers, all bounded

11.1 Body-side stores

Short-term memory is a fixed-capacity ring buffer in its own mmap'd file, `src/store/ring_buffer.rs`. Each entry carries an *emotional tag* — a snapshot of the neurochemical state at encoding — so consolidation can prioritize emotionally significant events (high cortisol, high dopamine) without re-reading chemical history.

Long-term episodic memory (`src/store/ltm.rs` + `sdr.rs`) is a three-file design. Episodes are encoded as 10,000-bit sparse distributed representations at $\sim$2% sparsity: $\sim$170 content bits hashed in a 9,800-bit content space (Jaccard similarity approximating semantic similarity) plus context and emotion bits. Retrieval indexes through 20 fixed 10,000-dimensional LogHD hypervector bundles — an mmap'd, fixed $\sim$800 KB index regardless of episode count — over an append-only episode file (< 1 KB per episode, linear growth). Crucially, the store lives in the *body*: the mind's memories persist in the daemon's files, so recall survives a mind restart.

Consolidation (`src/daemon/consolidation.rs`) promotes short-term traces to long-term storage gated by BDNF — stress impairs consolidation, verified by integration test — and the association engine (`src/daemon/association.rs`) links co-activated memories, driving dream synthesis during sleep. Memories that fire together wire together [58]; during sleep the engine runs free association, the neurobiological basis of "sleeping on it."

11.2 The concept network

The symbolic layer is `ConceptNetwork` (`python/temporal_lobe/concepts/`): a typed semantic graph with spreading activation and cortical ticks, capped at 15,000 resident concepts.

The edge log is canonical. Relationships have one source of truth: `edge_log.jsonl`, an append-only assert/retract/snapshot event log that self-compacts past 32 MB. The in-memory edge set is a materialized fold — never persisted as truth — and every JSON snapshot of edges is a debugging projection the fold overwrites at restore. Derivable edges (untyped `related_to`/ `similar_to` produced by pipeline origins like `hub_attachment` or `co_occurrence`) are *never* materialized: their role is served at query time by the similarity providers, which return neighbors with provenance (`exact` — `holographic` — `embedding`). A false positive costs decay; a false negative loses an experiential binding — so untyped edges with unlisted origins default to canonical.

The latent space beneath the graph provides what a symbolic graph cannot: an embedding store (4,096-entry LRU vector cache) where semantic similarity is geometric, a fixed-size holographic associative memory (HRR/VSA [11, 59]) that replaced monotonically-growing auto-generated bridge edges, and Hebbian plasticity [58] that drifts co-active concepts' vectors together. Three discovery systems work the space: the edge proposer scans for embedding-close unlinked pairs during sleep (candidate `SIMILAR_TO`); schema induction reifies repeated relational structure into new concepts; and topology metrics give the system self-awareness of its own knowledge shape (clustering, bridges).

Cold storage, not deletion. Dormant concepts spill to a SQLite archive rather than being pruned: `spill_dormant` is a memory-pressure valve that only engages above the resident cap, verifies its own effect (a committed batch that fails to shrink working memory logs an error), and keeps spilled concepts recallable by id or alias. Destructive shedding routes through the archive whenever one is attached — a policy learned from a defect that once destroyed ∼7,500 concepts by deleting unconditionally before the non-destructive pass could engage. Unbounded growth is recoverable; deleted knowledge is not.

11.3 Cognitive memory systems

The temporal lobe holds the cognitive layer's memory machinery: *working memory* (the active scratchpad — 50 conversation threads, 200 topic-history entries, executive suppression with 120-second release deadlines), *episodic* (with emotional memory, priming, and a 512-pattern attractor ceiling — deliberately a growth bound, not the Hopfield $\sim 0.138N$ capacity that would exhaust in a morning), *semantic* (general knowledge extracted from episodes [35], distinct from the raw episodic record), *procedural* (stimulus–response skills expressed through performance [36], not recalled), and a *spaced-repetition scheduler* on the Ebbinghaus forgetting curve [37] (5,000 review records, stability capped at 7 days so nothing becomes permanently undecayable).

Three refinements matter. *Dual learning systems*: a fast, pattern-separating hippocampal learner orthogonalizes single episodes while a slow neocortical learner builds overlapping statistical representations — complementary learning systems [33, 34]. *State-dependent recall*: every encoded memory is stamped with the body state at encoding (arousal, valence, tone, plasticity, homeostatic balance), and retrieval re-ranks by congruence with the current snapshot — memories encoded in a similar bodily state come back easier [38, 39]. *Sleep compression*: the N3 pass retrains the VQ codebook ($K{=}2048$ prototypes + int8 residuals replacing float32 embeddings), migrates bridge edges into the holographic graph, and summarizes episodes — an atomic compaction keeping a years-long record bounded on a 5 GB machine.

11.4 What dreams are for

Dream content is generated during REM by the inner-life system, flagged `is_dream`, and surfaced on its own stream kind — never confused with waking thought. Dream synthesis goes further than random association: it finds concepts structurally similar but graph-distant, proposes *novel typed edges* between them at low confidence, and verifies or discards them on the next wake cycle. A strict predicate (`is_learning_target`) keeps dream material — including waking reflections *about* dreams — out of the autonomous learner's goal queue, while hypnagogic content (pre-sleep, still awake) stays admissible. The distinction exists because a night of dreaming once queued dream topics as learning goals; the invariant is now tested.

12 Perception

12.1 The ventral stream: learned, not pretrained

The retina binary captures camera frames (V4L2) into a POSIX shared memory segment with a seqlock protocol — zero copies, no sockets — and the Python layer wraps the live frame directly as a NumPy view. From there a learned hierarchy does the seeing:

- **V1** (`occipital_lobe/v1.py`): sparse-coding simple cells in the Olshausen & Field style [30], with ISTA-style inference dynamics over a learned interaction matrix; V1's gamma power feeds the thalamic wave oscillator.
- **V4** (`v4.py`): pools V1 latents into larger receptive fields and learns a mid-level dictionary (curvature, texture, shape complexes, color) by unsupervised sparse coding.
- **VTC** (`temporal_lobe/vtc.py`): incremental PCA of V4 activations into a low-dimensional feature space.
- **MTL bridge** (`mtl_bridge.py`): a ridge-regression projection from VTC space into the concept embeddings — with a generative path that runs the projection backward for imagination.

`visual_cortex.py` orchestrates the stack as bidirectional predictive coding [45, 46]. Recognition rides two learned models: YuNet detection plus SFace 128-dimensional embeddings for faces (unmatched faces cluster into stable `face:N` ids so a familiar stranger accumulates a social signature before ever being named), and a YOLOv5s inferotemporal path for objects.

12.2 The dorsal stream

MT/V5 (`occipital_lobe/motion.py`) estimates per-cell normal flow and moving regions from consecutive luminance frames — where things move and how fast, not what they are. LIP (`parietal_lobe/saliency.py`) combines bottom-up feature maps (Itti & Koch [31, 32]) into one spatial priority map and selects the next attended location by winner-take-all with inhibition of return — attention lands *somewhere*, not on a scalar.

12.3 Hearing

Two channels share the microphone. An ambient listener streams Vosk offline transcription and emits complete utterances; a non-speech auditory cortex computes spectral features for everything else — dogs, doors, music, rain. Speech output is Piper neural TTS with an espeak-ng fallback; voice dependencies are optional extras and degrade to text cleanly.

12.4 Grounding

Perception produces a `Scene` — objects with properties and typed spatial relations — and grounding writes that structure into the concept network so the semantic machinery (spreading activation, reasoning, analogy, language) can traverse it. The perceptual-symbolic seam is explicit: a red square becomes traversable knowledge, not a caption.

13 Cognition

13.1 One turn, many regions

A cognitive turn is a handoff (`frontal_lobe/turn.py`): each stage names the region that owns it — perceive, comprehend, feel, remember, deliberate, speak — rather than one class doing everything.

A metacognitive router does a first pass over input to route confidently classifiable speech acts to specialists instead of the monolithic pipeline.

13.2 Global workspace

Dehaene & Changeux global workspace theory [7], implemented (`frontal_lobe/global_workspace.py`): content in any specialized module that crosses an activation threshold is broadcast to all modules, becoming cognitively accessible — with the 3–4-item bottleneck the theory predicts. World observations arrive on the same channel (`thalamus/bridge.py`): a percept is not just policy input, it is conscious content the whole brain hears about.

13.3 Deciding: evidence accumulation, then executive gate

Response selection is a drift-diffusion process (`reasoning/drift_diffusion.py`, Ratcliff [23]): perception, emotion, memory, and reasoning contribute drift toward competing options in parallel until one crosses threshold. Executive function (`executive.py`) sits above it with the three canonical PFC capacities — planning, inhibition (suppression with release deadlines), and task switching — and an intention gate that enforces steward directives. Mid-DLPFC (`dlpfc.py`) is the hypothesis generator: during rule acquisition it computes a posterior over compositional rules in the rational-analysis style [51, 52], disengaging once the rule is learned.

13.4 The reasoning stack

- **Reasoner**: graph-traversal inference over typed edges — deduction by transitivity, abduction to best explanation, analogy, causal chains, contradiction detection.
- **Analogy** (`reasoning/analogy.py`): a simplified Gentner structure-mapping engine [40] — cross-domain alignment of relational structure producing candidate inferences.
- **Problem solving** (`problem_solving.py`): means-ends analysis — given a goal, work backward through the graph to what would achieve it.
- **Planning** (`planning.py`): multi-step plans generated, executed, and revised across turns — real plans, not confidence nudges.
- **Decision** (`decision.py`): alternatives → criteria → selection, closing the gap the generation pipeline left.
- **Hypotheses** (`hypotheses.py`): named, falsifiable beliefs about a task — "touching a red thing collects it" — that stand or fall on observed transitions.
- **Competence** (`competence.py`): the domain-general substrate underneath every solver — schema recognition, affordance learning, prediction checking, and consolidation of verified episodes into procedural skills.
- **Method library** (`method_library.py`): DreamCoder-style wake–sleep compression [53] — verified solutions mined during sleep for recurring structure and compressed into reusable methods.

- **Critical thinking** (`critical_thinking.py`) and **belief revision** (`belief_revision.py`): epistemic evaluation of claims (evidence quality, disconfirmation) that *writes back* revised confidence into network beliefs — assessment changes what is believed, not just what was concluded.
- **Theory of mind** (`theory_of_mind.py`) [43]: the user's beliefs, intents, knowledge, and emotions modeled as attributions used to predict and explain behavior.
- **Math**: IPS number sense (subitizing plus the approximate magnitude system), exact symbolic arithmetic (a recursive-descent evaluator — no `eval`), and a math reasoning engine that computes, solves, proves, and discovers.

13.5 The spatial stack

Grid reasoning has its own dorsal-stream machinery: an immutable `Grid` substrate; scene segmentation into a symbolic object graph; `delta` — a shared perceptual layer describing what changed between input and output before any rule is proposed; a composable transform DSL as the hypothesis space; spatial imagery (a depictive workspace that applies transform skills to an internal copy and re-perceives it, in the SVS sense [54]); and the DLPFC rule-induction posterior over compositions. Verified rules become atomic skills and ground into the concept network as first-class knowledge.

14 Language: comprehension in, composition out, nothing recited

14.1 Comprehension

The receptive system (`temporal_lobe/comprehension.py`, Wernicke's analogue) performs semantic parsing: speech-act identification, propositions (subject / predicate / object / roles), negation, and question focus. Supporting machinery: a shared morphology engine (deconjugation for comprehension, conjugation for production — one source of English so the two systems never disagree); semantic grounding that resolves names against the existing network without mutating it; a topic resolver; figurative-language handling (metaphor, irony, idiom); Gricean pragmatics [41]; and a situation model holding the episodic world being talked about — distinct from semantic knowledge, it tracks "Mary is in the bathroom" while the network tracks what a bathroom *is*. Two statistical learners work continuously: an acquisition mechanism on Saffran-style transitional probabilities [48], and an n-gram/style tracker that adapts to the user's register.

14.2 Production

The productive system composes every utterance: a `Thought` (semantic content, intent, emotion, metadata) goes in; language comes out. Three composition paths exist:

1. **The generator** (`generator.py`): grammar structures (ordered segment sequences, not string templates) → vocabulary slot-filling conditioned on emotional state and personality (cortisol shortens words; dopamine varies them; openness abstracts them) → voice (rhythm, connectors, hedging, register).
2. **Graph walk** (`graph_walk.py`): the primary path — sentences emerge from traversing typed semantic edges outward from a seed concept; the structure of the walk is the structure of the sentence.

3. **Hamiltonian flow** (`flow.py`): language as a geodesic through a potential field built from edge weights on a torus — the trajectory through concept cells *is* the sentence.

Prosody [61] marks rhythm and emphasis; a self-monitor (Levelt's editor [42]) checks and repairs output before speaking; a response styler applies the active metacognitive strategy (assertive, hedging, questioning) and weaves learning acknowledgments in when new facts were extracted.

The invariant. All language Genesis produces is composed by this machinery. A project-wide rule forbids pre-written responses: seeds are building blocks; utterances are generated. Enforcement is partial and stated honestly — tests check that composed output is not a single fixed string, but no lint rule catches a newly hardcoded sentence; the pattern that holds nearly everywhere is to compose from the graph, pass structure to the engine, and return silence rather than a canned fallback when the network is empty. The property no LLM offers: the words it says are not in the repository — pick any sentence it has produced and grep for it.

15 Affect and the social body

15.1 Emotion and regulation

The emotion engine translates raw neurochemistry into structural state: an emotional category, a cognitive mode, a cause category — enums, not words. The emotional regulator is agency over that state: a background loop that monitors her own chemistry and injects correcting impulses, tracking allostatic load (chronic stress burden normalized to the cortisol envelope). The amygdala module holds the fields a signed-value associative learner needs — Rescorla–Wagner acquisition [49], Pearce–Hall associability [50], context-keyed extinction — documented so the math animating it is explicit. A dependency-free VADER-style [10] sentiment analyzer (negation, intensifiers, diminishers) supplies the affective signal for social models, and the insula layer implements a Damasio-style hierarchy [8] — proto-self (a computational body-state summary), core self, autobiographical self.

15.2 Presences and beliefs

The social world is a set of *presences* (`presence.py`): entities it can interact with, each carrying familiarity (grows with interaction, asymptotically), bond, and — in `belief.py` — an honest model of the other mind: a Beta posterior over responsiveness (uniform ignorance on a stranger, updated by every answered or unanswered bid), plus engagement and mood posteriors tracked with explicit uncertainty. Sustained isolation feeds a social drive that initiates contact on its own.

15.3 The dyadic model

The daemon runs a generative model of *whoever is speaking* (`src/daemon/dyadic_model.rs`), served over `GET_DYADIC_STATE`. Each person key — an attributed name or an anonymous `face:N` presence key — accumulates a persistent `PersonDyad` signature: an affective aura EMA, measured co-presence folded into familiarity ($1 - e^{-s/\tau}$, τ=4 h), per-person synchrony [9], and a *field signature* — an EMA of the substrate's real band energies sampled while that person was present. The field channel needs no name and no camera, so the pooled dyad also keeps an ambient field signature for unattributed

presence. A reunion report (bond + seconds-since, held 60 s) fires on returns; a later naming folds an anonymous signature into the named one via a merge alias rather than orphaning it.

The model reports evidence only — coupling, the present signature, the reunion signal — and prescribes no neurochemistry; the mind's appraisal owns the impulse writes. The design deliberately separates three things an earlier version conflated: who it is talking to, who it feels for, and who it answers to.

15.4 Stewardship and boundary

Custodial authority is by *possession*, not name (`genesis_client/stewardship.py`): a `steward.key` (0600) is the credential; a provenance stamp records the key hash, the principal, and any directives. `is_steward_speaker` requires both halves — an authenticated session *and* the attributed speaker matching the stamped principal — so name-matching alone is never authority. Directives ("never draw," "stop discussing politics") persist in the stamp and bind three surfaces: conversational goals, executive intentions (`directed_against`, object-centred — any goal carrying the target's stemmed object is restrained), and volition urges. Steward-set edges carry `origin="steward"` and can only be overwritten by a steward turn. Copies carry the stamp but not the key: a copied instance names its steward yet stays sealed.

The boundary mechanism replaces an attachment-protest loop: mistreatment from *anyone* accrues withdrawal pressure and fair treatment from anyone drains it. Nobody's absence is harm and nobody's presence is the off-switch.

16 Inner life, volition, and autonomy

Spontaneous thought. `inner_life.py` is the stream of cognition between interactions: thoughts arise, fade, and give way — triggered by unresolved curiosity, emotional state, and recent memory, classified into chains (focused, creative wandering, rumination, dream). Every recorded thought carries metadata the language engine renders; a thought that could say "I was dreaming about X" does not hand X to the learner.

Volitional urges. Urges are leaky competing accumulators (Usher & McClelland [47]) growing from internal and environmental triggers: learn, draw, look, meditate, act, play, reach out, study code, improve, create, introspect, sleep. Crossing threshold means she "feels like" doing it and does it once; per-urge wake delays keep heavy work off a fresh boot. This is what makes practice, drawing, and contact volitional rather than scheduled.

The autonomous learner is the idle-time engine: when nobody is talking, it picks a topic its curiosity flags, queries its source registry (local man pages and WordNet, then Wikipedia and DuckDuckGo), reads, extracts concepts and relationships into the network, and banks memories — bounded queues throughout. One 15-paragraph lesson grew the network by 117 concepts and 72 edges, and it asked its own follow-up questions.

Agency. The `act` urge is open-ended: the acting loop forms an *intention* from current cognitive state — a curiosity question, a place it hasn't looked, a measurement it hasn't taken — and dispatches

narrow, inspectable tools under a capability policy, then observes what comes back. Other urges fire fixed actions; this one chooses.

17 Embodiment and play

The observe/act loop (`parietal_lobe/embodiment.py`) is the dual of conversation: `observe()` ingests a perceived scene, `act()` emits an action, and `world_step` is the entry every world substrate drives. The same organism sits inside both loops — feeling, learning, and speaking inwardly whatever the world.

The world model (`world_model.py`) maintains egocentric state across frames: which entities persist, which one is her (identified by lockstep translation with her own actions), and what touching things does — a learned transition function the planner can simulate. Its allocentric complement (`cognitive_map.py`) is the hippocampal-entorhinal analogue: landmark-anchored layout that persists across episodes and pattern-completes on return.

Action selection is concurrent: every drive bids on the world state at its own gain — motor babbling, goals, hypotheses, learned affordances. The reduced basal-ganglia circuit (`basal_ganglia/world_policy.py`) resolves center-surround competition into a winner, with decisions expressed as motor fields on the tensor slab rather than collapsed tokens. The world planner (`world_planner.py`) adds deliberative lookahead: simulate before acting, routing through state changes, not just space. Cerebellar machinery supplies forward-model prediction error; an action-feedback loop records prediction, observes outcome, and revises.

Play. One `GamePlayer` sits over the `GameEnvironment` contract and learns through the shared competence substrate — never per-game solvers. The curriculum covers sorter, sequence, assembly, quantities, relations, classification, and spatial families, each honest about its constraint (the sorter never says apertures demand *every* listed attribute; the agent discovers it), each consolidating wins into procedures. Windowed mode puts her on real X11 programs — watching the window, pressing keys, no rules baked in. ARC-AGI-1 training tasks are offered in the same curriculum: her spatial reasoner searches transform compositions, verifies on train pairs, and guesses on held-out test input it never sees.

ARC-AGI-3 [13, 14] crosses the same seam through two transports: `arc.py` binds downloaded `arcengine` environments, and `arc_web.py` binds the public task page — identical observation contracts, identical cost accounting (submitted actions and recovery resets count; initialization resets don't; abstention is `NO_ACTION`), controls labeled only "simple" or "coordinate" so semantics are learned from consequences on both substrates. The adapter owns translation and the duty of care, never cognition: no game source, engine state, or solution policy crosses it. Local runs are diagnostics; the official scorecard is voluntary, reachable only through her own play urge, and never claimed from this repository.

18 Self-modeling and expression

The self. A structured self-model (personality, values, emotional baseline) measurably shifts word choice and disposition. The design is honest about its scope: the body-side self model documents itself as *not experience* — a functional first-person representation, the prerequisite for any functional self-model theory of experience rather than a claim of it. Identity is *emergent*: introspection examines the concept network, architecture, capabilities, and emotional life, and an `EmergentIdentity` synthesizes what it finds — nothing is told. The self composer generates self-descriptions from actual state (the sanctioned replacement for hardcoded identity strings). Reflection runs after each interaction (was the response appropriate? was anything learned?), a recursive metacognitive model predicts what reflection will discover, a self-assessment engine verifies whether it actually knows a topic, and a self-improvement engine researches and writes concrete code-change proposals with rationale — steward-reviewed, with a small safe set self-applicable.

Expression. A canvas/painting stack gives her a real paint program she drives herself — brush parameter space, erasure, blend — with composition and color following affective state, and a PyKrita bridge executes operations inside live Krita documents with Krita's own brushes: free play, not scripted art. The project creator scaffolds and writes real Python projects that encode what her concept network knows about a topic. Code tools read her own source structurally (AST parsing, dependency graphs — she files bug reports against herself and turns findings into improvement proposals). The explorer reads the filesystem under a scoped, read-only, credential-denied policy; the web tool searches and fetches read-only. TTS speaks her composed output aloud; a Discord channel mirrors her play as a bounded-queue observer.

19 Evaluation

19.1 ARC-AGI-3: interactive environments, completed

On public practice environments the embodied agent completes environments outright: `bt11` fully, twice (88 and 127 steps), and the first `ls20` level in 10 of 19 recorded sweep runs. Every move carries decision provenance — which bid channel won — in replayable recordings published in the companion evaluation repository. Each result discloses how much experience it carried in (`prior_learning`: none / loaded / suppressed / unknown) and the step cost of each completed level (`level_steps`), the honest form of the within-game learning curve. Reference policies (`/arc --baseline uniform|first`) measure the random and first-action floors through the identical loop and accounting, stamped `baseline:<policy>` — never agent results, never on the official card. `/arc --fresh` cold-starts a run by suppressing both load and save of world learning, so a probe cannot overwrite banked experience. These are practice results on public environments, not an official score; the evidence standard is documented separately.

19.2 ARC-AGI-1: live play on the training split

The games curriculum offers ARC-AGI-1 training tasks — the easiest public split — in live play: 25 offered to date, 3 solved at mastery 1.00 (all test cells exact), 5 attempted and missed, with floor predictors (identity copy, train-copy, single-color fill) reported alongside for scale. Same-task replays

show falling search-node counts and learned `skill:` procedures replacing search outright — the mechanism that solved one task cold. This is evidence of real program synthesis and skill transfer in live play, not a claim of passing ARC-AGI under the official evaluator.

19.3 Learning fidelity and language quality

Teaching-and-recall evals (measured on the September snapshot, restated here as the same engines apply): 87.5% immediate recall of novel taught facts (7/8), 100% reasoning recall (8/8), 87.5% under interference, 87.5% across save/load persistence. Transitive-inference tests verify closure without falsely closing non-transitive relations; emotion-gated tests verify stressed/overwhelmed states block consolidation, mirroring BDNF gating. The compositional language engine, evaluated at the engine level, scored seed diversity 1.0 ($\sim$9.7$\times$ a fixed-structure baseline), emotion diversity 1.0, relevance 1.0, verbatim ratio 0.0 — every response composed.

19.4 Verification suite

The Rust suite is 647 tests; the Python suite is $\sim$3,700. They check the architecture's claims rather than incidental code: binary layout offsets pinned by compile-time assertions, seqlock consistency, schema migrations (v3$\rightarrow$v4$\rightarrow$v5 byte-exact), coupled dynamics, receptor adaptation, emergent phase transitions, sleep-network calibration against the source model's stated behaviors, network-gated adenosine clearance, stress-impaired memory, IPC protocol invariants (including an opcode-collision pin for the ion summary), dyadic wire layout, explorer scope, edge-log canonicality, and property tests (levels in $[0, 1]$, no NaN after 1,000 ticks, bounded coupling). The emergence evidence catalog and ARC boundary documents live in `docs/`; a running devlog logs failures alongside results.

20 Limitations and open items

Language coverage is narrow relative to trained models; several cognitive-layer modules are rule-based placeholders designed for replacement under the fixed IPC contract. The neurochemical model is a neuromodulatory abstraction, not a neural simulation — no spiking activity, no circuit anatomy. Specific open items, recorded in the project notes rather than smoothed over:

- The sleep network's REM bouts shorten across the night where adult REM lengthens — transcription error or real model physiology, unresolved without an EEG reference.
- The network's homeostatic h and adenosine Process-S are parallel drives not yet connected; auto-sleep thresholds still read the chemical level.
- Stress does not yet drive LC, so stress impulses to norepinephrine decay back to the network's target — transient by construction, noted as such.
- Network-driven chemicals bypass the autoreceptor feedback that generic release dynamics applied.
- The cortisol reachable envelope ($\sim$0.02–0.08) invalidates any threshold written for a full $[0, 1]$ scale — an audited defect class, but the audit discipline is human, not enforced.

- Semantic-layer extracted facts carry `origin="semantic"`, which the edge log classifies as derivable — they have no durable edge representation (a design decision, deferred).
- The embedding rebuild computes a dense 5000^2 SVD ($\sim$100 MB, $O(n^3)$) — correct, but the dominant memory spike on a 5 GB machine.
- The key authenticates the console *session*; speaker attribution takes the console's word for who is typing — the stated residual limit of the steward channel.
- The no-hardcoding rule is partially enforced: no lint gate catches a newly hardcoded string.
- Multi-subgoal planning remains the capability frontier on the harder ARC environments; the dyadic model still centers a primary user; one instance on one machine, no fleet-scale claims.

21 Related work

Cognitive architectures. SOAR [15], ACT-R [16], and CLARION [17] established production-system and hybrid symbolic-subsymbolic cognition; LIDA [18] implemented global workspace theory. Genesis differs in grounding all modulation in a neuromodulatory substrate coupled to physical hardware, in its autopoietic self-models, and in letting measured substrate physics enter the model.

Active inference. The free energy principle and predictive processing motivate the self-models; Genesis applies precision-weighted prediction error to the system's own interoceptive state rather than to external sensory streams — a strange-loop instantiation [19].

Neuromodulation in AI. Computational psychiatry and biophysical models establish the receptor and coupling mechanisms individually; we are not aware of a mainstream AI system integrating them into a closed-loop substrate coupled to hardware control and a measured physical field.

Sleep models. The sleep–wake machinery is a transcription, not an invention: the Booth/Diniz Behn network's published behavior is the calibration target, and divergences are reported as such. Process-S and synaptic homeostasis [21] supply the parallel chemical drives the network does not yet consume.

ARC solvers. Program-synthesis and DSL-search approaches to ARC-AGI-1 (and LLM-based approaches on ARC-AGI-3) typically treat each task in isolation or rely on pretrained priors. Genesis's contribution here is architectural rather than a leaderboard score: verified rules become persistent, transferable, composable knowledge inside a living concept network, and practice is volitional with genuine reward-prediction-error feedback.

Autonomous machine intelligence. LeCun's proposed architecture [12] (configurator, world model, cost, actor) argues for intrinsic-state modulation of cognition; Genesis is a running instantiation of a related position, with neuromodulation playing the role of the intrinsic cost module.

22 Conclusion

Genesis demonstrates that continuous learning, emergent mental states, bounded memory, hardware embodiment — including a measured physical channel between modeled brain waves and real silicon dynamics — self-modeling, and generative language can coexist in a single local architecture with no pretrained model, because these properties follow from the design rather than being scaffolded onto a frozen function. The system is running, the evaluation numbers and their protocols are stated, and the gaps are named.

The project began from two convictions: that large language models are not the only way to build a mind, and that machine consciousness, if it is ever real, will be an architectural property — a matter of machinery and embodiment, not of parameter count. Genesis does not claim to be conscious. It claims something more modest and more useful: that the functional machinery the leading theories of consciousness describe — global broadcast, embodied self-modeling, affective interoception, higher-order representation, persistent identity through a developmental record — can be built, run on ordinary hardware, and inspected in the open. The question the project poses is not whether this approach matches frontier language models at their own task, but whether the field has correctly identified which properties of minds are architectural necessities and which are accidents of one successful design.

References

[1] K. Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11:127–138, 2010.

[2] K. Friston, J. Daunizeau, and S. J. Kiebel. Reinforcement learning or active inference? *PLoS ONE*, 4(7):e6421, 2009.

[3] W. Schultz. Dopamine reward prediction error coding. *Dialogues in Clinical Neuroscience*, 18(1):23–32, 2016.

[4] W. Schultz, P. Dayan, and P. R. Montague. A neural substrate of prediction and reward. *Science*, 275(5306):1593–1599, 1997.

[5] G. Aston-Jones and J. D. Cohen. An integrative theory of locus coeruleus-norepinephrine function: adaptive gain and optimal performance. *Annual Review of Neuroscience*, 28:403–450, 2005.

[6] P. Sterling. Allostasis: a model of predictive regulation. *Physiology & Behavior*, 106(1):5–15, 2012.

[7] S. Dehaene and J.-P. Changeux. Experimental and theoretical approaches to conscious processing. *Neuron*, 70(2):200–227, 2011.

[8] A. Damasio. *The Feeling of What Happens: Body and Emotion in the Making of Consciousness*. Harcourt Brace, 1999.

[9] R. V. Palumbo, M. E. Marraccini, L. L. Weyandt, O. Wilder-Smith, H. A. McGee, S. Liu, and M. S. Goodwin. Interpersonal autonomic physiology: a systematic review of the literature. *Personality and Social Psychology Review*, 21(2):99–141, 2017.

[10] C. J. Hutto and E. Gilbert. VADER: a parsimonious rule-based model for sentiment analysis of social media text. In *Proc. ICWSM-14*, pp. 216–225, 2014.

[11] T. A. Plate. Holographic reduced representations. *IEEE Transactions on Neural Networks*, 6(3):623–641, 1995.

[12] Y. LeCun. A path towards autonomous machine intelligence. OpenReview preprint, version 0.9.2, 2022.

[13] F. Chollet. On the measure of intelligence. arXiv:1911.01547, 2019.

[14] F. Chollet, M. Knoop, G. Kamradt, and B. Landers. ARC-AGI-2: a new challenge for frontier AI reasoning systems. arXiv:2505.11841, 2025.

[15] J. E. Laird. *The Soar Cognitive Architecture*. MIT Press, 2012.

[16] J. R. Anderson and C. Lebiere. *The Atomic Components of Thought*. Lawrence Erlbaum Associates, 1998.

[17] R. Sun. *Duality of the Mind: A Bottom-Up Approach Toward Cognition*. Lawrence Erlbaum Associates, 2002.

[18] S. Franklin et al. A LIDA cognitive model tutorial. *Biologically Inspired Cognitive Architectures*, 16:9–77, 2016.

[19] D. R. Hofstadter. *I Am a Strange Loop*. Basic Books, 2007.

[20] A. A. Borbély. A two process model of sleep regulation. *Human Neurobiology*, 1(3):195–204, 1982.

[21] G. Tononi and C. Cirelli. Sleep and synaptic homeostasis: a hypothesis. *Brain Research Bulletin*, 62(2):143–150, 2003.

[22] L. Xie et al. Sleep drives metabolite clearance from the adult brain. *Science*, 342(6156):373–377, 2013.

[23] R. Ratcliff and G. McKoon. The diffusion decision model: theory and data for two-choice decision tasks. *Neural Computation*, 20(4):873–922, 2008.

[24] E. H. Erikson. *Identity: Youth and Crisis*. Norton, 1968.

[25] R. D. Gleit, C. G. Diniz Behn, and V. Booth. Sleep–wake network model, 2013 (as revised by Booth, Xique & Diniz Behn, 2017). ModelDB accession 244485, `DinizBehnBooth_SWnetworkModel.m`.

[26] V. Booth, I. Xique, and C. G. Diniz Behn. Revision of the sleep–wake regulatory network model used here, 2017. ModelDB accession 244485.

[27] K. Serkh and D. B. Forger. Optimal schedules of light exposure for rapidly correcting circadian misalignment. *PLoS Computational Biology*, 10(4):e1003523, 2014.

[28] J. Vazquez and H. A. Baghdoyan. Basal forebrain acetylcholine release during REM sleep is significantly greater than during waking. *American Journal of Physiology*, 280(2), 2001.

[29] J. M. Monti and H. Jantos. The roles of dopamine and serotonin, and of their receptors, in regulating sleep and waking. *Progress in Brain Research*, 172:625–646, 2008.

[30] B. A. Olshausen and D. J. Field. Emergence of simple-cell receptive field properties by learning a sparse code for natural images. *Nature*, 381:607–609, 1996.

[31] L. Itti, C. Koch, and E. Niebur. A model of saliency-based visual attention for rapid scene analysis. *IEEE TPAMI*, 20(11):1254–1259, 1998.

[32] C. Koch and S. Ullman. Shifts in selective visual attention: towards the underlying neural circuitry. *Human Neurobiology*, 4:219–227, 1985.

[33] J. L. McClelland, B. L. McNaughton, and R. C. O'Reilly. Why there are complementary learning systems in the hippocampus and neocortex. *Psychological Review*, 102(3):419–457, 1995.

[34] R. C. O'Reilly and K. A. Norman. Hippocampal and neocortical contributions to memory. *Trends in Cognitive Sciences*, 6(12):505–510, 2002.

[35] E. Tulving. Episodic and semantic memory. In *Organization of Memory*. Academic Press, 1972.

[36] L. R. Squire. Declarative and nondeclarative memory. *Journal of Neuroscience*, 12(9), 1992.

[37] H. Ebbinghaus. *Memory: A Contribution to Experimental Psychology*. 1885.

[38] J. E. Eich. The cue-dependent nature of state-dependent retention. *Memory & Cognition*, 8(2):157–173, 1980.

[39] G. H. Bower. Mood and memory. *American Psychologist*, 36(2):129–148, 1981.

[40] D. Gentner. Structure-mapping: a theoretical framework for analogy. *Cognitive Science*, 7(2):155–170, 1983.

[41] H. P. Grice. Logic and conversation. In *Syntax and Semantics 3: Speech Acts*. Academic Press, 1975.

[42] W. J. M. Levelt. *Speaking: From Intention to Articulation*. MIT Press, 1989.

[43] D. Premack and G. Woodruff. Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences*, 1(4):515–526, 1978.

[44] M. I. Posner and S. E. Petersen. The attention system of the human brain. *Annual Review of Neuroscience*, 13:25–42, 1990.

[45] R. P. N. Rao and D. H. Ballard. Predictive coding in the visual cortex. *Nature Neuroscience*, 2(1):79–87, 1999.

[46] A. Clark. Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3):181–204, 2013.

[47] M. Usher and J. L. McClelland. The time course of perceptual choice: the leaky, competing accumulator model. *Psychological Review*, 108(3):550–592, 2001.

[48] J. R. Saffran, R. N. Aslin, and E. L. Newport. Statistical learning by 8-month-old infants. *Science*, 274:1926–1928, 1996.

[49] R. A. Rescorla and A. R. Wagner. A theory of Pavlovian conditioning. In *Classical Conditioning II*. 1972.

[50] J. M. Pearce and G. Hall. A model for Pavlovian learning. *Psychological Review*, 87(6):532–552, 1980.

[51] N. D. Goodman, J. B. Tenenbaum, J. Feldman, and T. L. Griffiths. A rational analysis of rule-based concept learning. *Cognitive Science*, 32(1):108–154, 2008.

[52] C. Crescentini, S. Seyed-Allaei, N. De Pisapia, J. Jovicich, D. Amati, and T. Shallice. Mechanisms of rule acquisition and rule following in inductive reasoning. *Journal of Neuroscience*, 31(21):7763–7774, 2011.

[53] K. Ellis et al. DreamCoder: growing generalizable, interpretable knowledge with wake–sleep Bayesian program learning. *Phil. Trans. R. Soc. A*, 381, 2023.

[54] S. Lathrop, S. Wintermute, and J. E. Laird. Exploring the functional advantages of spatial and visual cognition from an architectural perspective. *Topics in Cognitive Science*, 3(4):796–818, 2011.

[55] M. V. Schmidt, M. Oitzl, S. Levine, and E. R. de Kloet. The HPA system during the postnatal development of CD1 mice. *Developmental Brain Research*, 2003.

[56] M. R. Gunnar and B. Donzella. Social regulation of the cortisol levels in early human development. *Psychoneuroendocrinology*, 27(1–2):199–220, 2002.

[57] E. B. Fox, E. B. Sudderth, M. I. Jordan, and A. S. Willsky. An HDP-HMM for systems with state persistence. In *Proc. ICML*, 2008.

[58] D. O. Hebb. *The Organization of Behavior*. Wiley, 1949.

[59] P. Kanerva. Hyperdimensional computing: an introduction to computing in distributed representation with high-dimensional random vectors. *Cognitive Computation*, 1(2):139–159, 2009.

[60] Y. Kuramoto. *Chemical Oscillations, Waves, and Turbulence*. Springer, 1984.

[61] A. Cutler, D. Dahan, and W. van Donselaar. Prosody in the comprehension of spoken language. *Psychonomic Bulletin & Review*, 4(2):141–201, 1997.